Fluid Mechanics ME 21201 Quiz 1

August 7, 2026

1. (1 pt) For a Newtonian fluid, which of the following statements is true?

   A. Shear stress is proportional to shear strain
   B. Rate of shear stress is proportional to shear strain
   C. Shear stress is proportional to the rate of shear strain
   D. Rate of shear stress is proportional to rate of shear strain

   Shear stress is proportional to the rate of shear strain

2. (1 pt) The continuum assumption in fluid mechanics implies that

   A. Fluid properties vary drastically in space.
   B. Fluid properties vary continuously in space.
   C. Fluids are composed of molecules.
   D. Flow velocities are discontinuous functions of space.

   Fluid properties vary continuously in space.

3. (1 pt) A solid sphere of diameter $d = 6$ mm is rising through oil ($\rho_o = 900$ kg/m^3, $\mu_0 = 0.7$ kg/ms) at a constant velocity of $v = 1$ cm/s (Assume that the sphere has reached a terminal velocity, such that the force due to viscosity opposes the direction of motion, and the magnitude is equal to $3\pi\mu vd$). What is the specific weight of the material from which the sphere is made? Take $g = 9.81$ ms^{-2}.

   A. 4.3 kN m^{-3}
   B. 5.3 kN m^{-3}
   C. 8.7 kN m^{-3}
   D. 12.3 kN m^{-3}

   5.3 kN m^{-3}

4. (1 pt) A solid wood block floats in a bucket of water with half of its volume submerged. If this entire setup is placed inside an elevator accelerating upward, what happens to the submerged fraction of the block?

   A. It increases because the water gets heavier.
   B. It decreases because the buoyant force increases.
   C. It remains exactly the same.
   D. It depends on the exact value of the acceleration.

   It remains exactly the same. The submerged volume fraction of a floating body depends on the ratio of the object's density to the fluid's density. When the elevator accelerates upward, the effective acceleration due to gravity changes to $g + a$. This increases both the block's weight and the buoyant force by the same scaling factor, leaving the submerged fraction unchanged.

5. (1 pt) A closed container completely filled with water is accelerated horizontally to the right. At which location inside the container is the fluid pressure the highest?

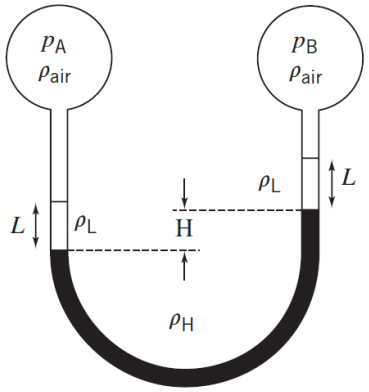


Figure 1: Figure for problem 6

A. Top-right corner
B. Top-left corner
C. Bottom-right corner
D. Bottom-left corner

Bottom-left corner.

6. (2 pts.) For the manometer setup shown in the figure, the pressure difference $p_A - p_B$ is given by

A. $(\rho_H - \rho_{\text{air}})gH$
B. $(\rho_H - \rho_L)gH$
C. $(\rho_H - \rho_L)gH + (\rho_L - \rho_{\text{air}})g(L - H)$
D. $(\rho_H - \rho_L)gL + (\rho_L - \rho_{\text{air}})gH$

$(\rho_H - \rho_{\text{air}})gH$

7. (1 pt) For a completely submerged, vertical flat surface, the center of pressure is always located _____ the centroid of the surface.
below

8. (2 pts.) A rectangular plate of width w and depth d is submerged vertically in a fluid with its top edge exactly at the water surface. What is the total hydrostatic force on one side of the plate? (Assume fluid density is ρ, gravity is g, and neglect atmospheric pressure). _____
$\frac{1}{2}\rho g w d^2$